

Title: Design and develop a distributed application to find the coolest/hottest year from the available weather data. Use weather data from the Internet and process it using MapReduce.

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ubuntu-406@ubuntu-406-OptiPlex-3000:~/log-mapreduce$ rm -rf output
ubuntu-406@ubuntu-406-OptiPlex-3000:~/log-mapreduce$ mkdir ~/weather_log-mapreduce
ubuntu-406@ubuntu-406-OptiPlex-3000:~/log-mapreduce$ cd ~/weather_log-mapreduce
ubuntu-406@ubuntu-406-OptiPlex-3000:~/weather_log-mapreduce$ nano sample_weather.txt
ubuntu-406@ubuntu-406-OptiPlex-3000:~/weather_log-mapreduce$ nano mapper.py
ubuntu-406@ubuntu-406-OptiPlex-3000:~/weather_log-mapreduce$ nano reducer.py
ubuntu-406@ubuntu-406-OptiPlex-3000:~/weather_log-mapreduce$ chmod +x mapper.py reducer.py
ubuntu-406@ubuntu-406-OptiPlex-3000:~/weather_log-mapreduce$ hadoop jar $HADOOP_HOME/share/hadoop/tools/lib/hadoop-streaming-*.jar \
-input sample_weather.txt \
-output output \
-mapper mapper.py \
-reducer reducer.py
2026-04-13 15:31:25,922 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-04-13 15:31:25,964 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-04-13 15:31:25,964 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-04-13 15:31:25,971 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-04-13 15:31:26,045 INFO mapred.FileInputFormat: Total input files to process : 1
2026-04-13 15:31:26,053 INFO mapreduce.JobSubmitter: number of splits:1
2026-04-13 15:31:26,124 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local892571456_0001
2026-04-13 15:31:26,124 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-04-13 15:31:26,180 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-04-13 15:31:26,182 INFO mapreduce.Job: Running job: job_local892571456_0001
2026-04-13 15:31:26,182 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-04-13 15:31:26,184 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-04-13 15:31:26,187 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-13 15:31:26,187 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-13 15:31:26,202 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-04-13 15:31:26,204 INFO mapred.LocalJobRunner: Starting task: attempt_local892571456_0001_m_000000_0
2026-04-13 15:31:26,215 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-04-13 15:31:26,215 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-04-13 15:31:26,222 INFO mapred.Task: Using ResourceCalculatorProcessTree : [ ]
2026-04-13 15:31:26,226 INFO mapred.MapTask: Processing split: file:/home/ubuntu-406/weather_log-mapreduce/sample_weather.txt:0+260
2026-04-13 15:31:26,231 INFO mapred.MapTask: numReduceTasks: 1
2026-04-13 15:31:26,269 INFO mapred.MapTask: (EQUATOR) 0 kvi 26214396(104857584)
2026-04-13 15:31:26,269 INFO mapred.MapTask: mapreduce.task.io.sort.mb: 100

FILE: Number of large read operations=0
FILE: Number of write operations=0
Map-Reduce Framework
  Map input records=10
  Map output records=30
  Map output bytes=230
  Map output materialized bytes=296
  Input split bytes=114
  Combine input records=0
  Combine output records=0
  Reduce input groups=3
  Reduce shuffle bytes=296
  Reduce input records=30
  Reduce output records=3
  Spilled Records=60
  Shuffled Maps =1
  Failed Shuffles=0
  Merged Map outputs=1
  GC time elapsed (ms)=4
  Total committed heap usage (bytes)=520093696
Shuffle Errors
  BAD_ID=0
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=260
File Output Format Counters
  Bytes Written=53
2026-04-13 15:31:27,202 INFO streaming.StreamJob: Output directory: output
ubuntu-406@ubuntu-406-OptiPlex-3000:~/weather_log-mapreduce$ cat output/part-00000
DEW_avg 23.5
TEMP_avg 33.5
WIND_avg 19.4
```